

```
#libraries
import pandas as pd
# used for data manipulation(add,delete,remove...)
```

```
df=pd.read_csv('/content/Hotel Reservations.csv.zip')
df.head() #to see the top records from above data #df can be any name
```

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space
0	INN00001	2	0	1	2	Meal Plan 1	0
1	INN00002	2	0	2	3	Not Selected	0
2	INN00003	1	0	2	1	Meal Plan 1	0
3	INN00004	2	0	0	2	Meal Plan 1	0
4	INN00005	2	0	1	1	Not Selected	0

data structures libraries in python how to work with data using libraries(manipulation) data analysis

```
df.shape
```

```
(36275, 19)
```

```
df['type_of_meal_plan']
```

	type_of_meal_plan
0	Meal Plan 1
1	Not Selected
2	Meal Plan 1
3	Meal Plan 1
4	Not Selected
...	...
36270	Meal Plan 1
36271	Meal Plan 1
36272	Meal Plan 1
36273	Not Selected
36274	Meal Plan 1

36275 rows × 1 columns

dtype: object

```
df['type_of_meal_plan'].apply(lambda i: i.replace('1','0'))
```

	type_of_meal_plan
0	Meal Plan 0
1	Not Selected
2	Meal Plan 0
3	Meal Plan 0
4	Not Selected
...	...
36270	Meal Plan 0
36271	Meal Plan 0
36272	Meal Plan 0
36273	Not Selected
36274	Meal Plan 0

36275 rows × 1 columns

dtype: object

```
for i in df['type_of_meal_plan']:  
    i= i.replace('1','0')  
    print(i)
```

Meal Plan 0

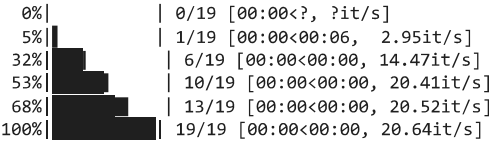
Meal Plan 0
Meal Plan 0
Meal Plan 0
Meal Plan 0
Meal Plan 0
Meal Plan 0

```
from ydata_profiling import ProfileReport
df= pd.DataFrame(df)
profile=ProfileReport(df, title="Employee Data Analysis")
```

[Upgrade to ydata-sdk](#)
Improve your data and profiling with ydata-sdk, featuring data quality scoring, redundancy detection, outlier identification, text validation, and synthetic data generation.

profile

Summarize dataset: 100% 128/128 [00:15<00:00, 5.15it/s, Completed]



Generate report structure: 100% 1/1 [00:04<00:00, 4.98s/it]

Render HTML: 100% 1/1 [00:00<00:00, 1.41it/s]

Employee Data Analysis Overview Variables Interactions Correlations Missing values Sample

Overview

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Overview Alerts 16 Reproduction

Dataset statistics

Number of variables	19
Number of observations	36275
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	5.3 MiB
Average record size in memory	152.0 B

Variable types

Text	1
Categorical	8
Numeric	10

Variables

Select Columns

Start coding or [generate](#) with AI.

```
#libraries
#Pandas(manipulation,analysis,reading data)
```

```
#numpy
#data analysis, manipulation
```

```
#numpy(numerical data)-when we work with large no.s of data
```

```
import numpy as np
arr=np.array([1,2,3,4,5]) #array=matrix
print(arr)
print(type(arr))
```

```
[1 2 3 4 5]
<class 'numpy.ndarray'>
```

```
arr=np.array([[1,2,3],[4,5,6]])
arr*2
```

```
array([[ 2,  4,  6],
       [ 8, 10, 12]])
```

```
import pandas as pd
from ydata_profiling import ProfileReport
jaan=pd.read_csv('/content/loan_data.csv')
jaan.head()
```



	person_age	person_gender	person_education	person_income	person_emp_exp	person_home_ownership	loan_amnt	loan_intent	loan_
0	22.0	female	Master	71948.0	0	RENT	35000.0	PERSONAL	
1	21.0	female	High School	12282.0	0	OWN	1000.0	EDUCATION	
2	25.0	female	High School	12438.0	3	MORTGAGE	5500.0	MEDICAL	
3	23.0	female	Bachelor	79753.0	0	RENT	35000.0	MEDICAL	
4	24.0	male	Master	66135.0	1	RENT	35000.0	MEDICAL	

```
analysis= ProfileReport(jaan)
```

```
analysis
```

Summarize dataset: 100%87/87 [00:12<00:00, 5.82it/s, Completed]

0%| | 0/14 [00:00<?, ?it/s]

7%|█| 1/14 [00:00<00:01, 6.70it/s]

21%|██| 3/14 [00:00<00:01, 8.87it/s]

29%|███| 4/14 [00:00<00:01, 8.90it/s]

43%|████| 6/14 [00:00<00:00, 9.11it/s]

57%|█████| 8/14 [00:00<00:00, 9.90it/s]

79%|██████| 11/14 [00:00<00:00, 14.17it/s]

100%|██████████| 14/14 [00:01<00:00, 12.23it/s]

Generate report structure: 100%1/1 [00:03<00:00, 3.66s/it]

Render HTML: 100%1/1 [00:00<00:00, 3.30it/s]

YData Profiling Report

OverviewVariablesInteractionsCorrelationsMissing valuesSample

Overview

Brought to you by YData

OverviewAlerts9Reproduction

Dataset statistics

Number of variables	14
Number of observations	45000
Missing cells	0
Missing cells (%)	0.0%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	4.8 MiB
Average record size in memory	112.0 B

Variable types

Numeric	8
Categorical	5
Boolean	1

Variables

Select Columns

```
import matplotlib.pyplot as plt
import seaborn as sns #data visualisation
```

```
print('/nData Types:/n',df.dtypes)
```

```
/nData Types:/n Booking_ID          object
no_of_adults                       int64
no_of_children                     int64
no_of_weekend_nights                int64
no_of_week_nights                   int64
type_of_meal_plan                   object
required_car_parking_space          int64
room_type_reserved                  object
lead_time                           int64
arrival_year                        int64
arrival_month                       int64
arrival_date                        int64
market_segment_type                 object
repeated_guest                      int64
no_of_previous_cancellations        int64
no_of_previous_bookings_not_canceled int64
avg_price_per_room                  float64
no_of_special_requests              int64
booking_status                      object
```

https://colab.research.google.com/drive/1cnEK45avMTe7vwgGU3Ca06ZqpglS5EwR#printMode=true

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```
dtype: object
```

```
print('\nMissing Values:\n',df.isnull().sum())
```

```
Missing Values:
Booking_ID                0
no_of_adults              0
no_of_children            0
no_of_weekend_nights      0
no_of_week_nights         0
type_of_meal_plan         0
required_car_parking_space 0
room_type_reserved        0
lead_time                 0
arrival_year              0
arrival_month             0
arrival_date              0
market_segment_type       0
repeated_guest            0
no_of_previous_cancellations 0
no_of_previous_bookings_not_canceled 0
avg_price_per_room        0
no_of_special_requests    0
booking_status            0
dtype: int64
```



```
print('\nSummary Statistics:\n',df.describe(include='all'))
```

```
n\SSummary Statistics:
Booking_ID  no_of_adults  no_of_children  no_of_weekend_nights  \
count      36275    36275.000000    36275.000000    36275.000000
unique      36275         NaN         NaN         NaN
top      INN36275         NaN         NaN         NaN
freq         1         NaN         NaN         NaN
mean         NaN     1.844962     0.105279     0.810724
std         NaN     0.518715     0.402648     0.870644
min         NaN     0.000000     0.000000     0.000000
25%         NaN     2.000000     0.000000     0.000000
50%         NaN     2.000000     0.000000     1.000000
75%         NaN     2.000000     0.000000     2.000000
max         NaN     4.000000    10.000000     7.000000

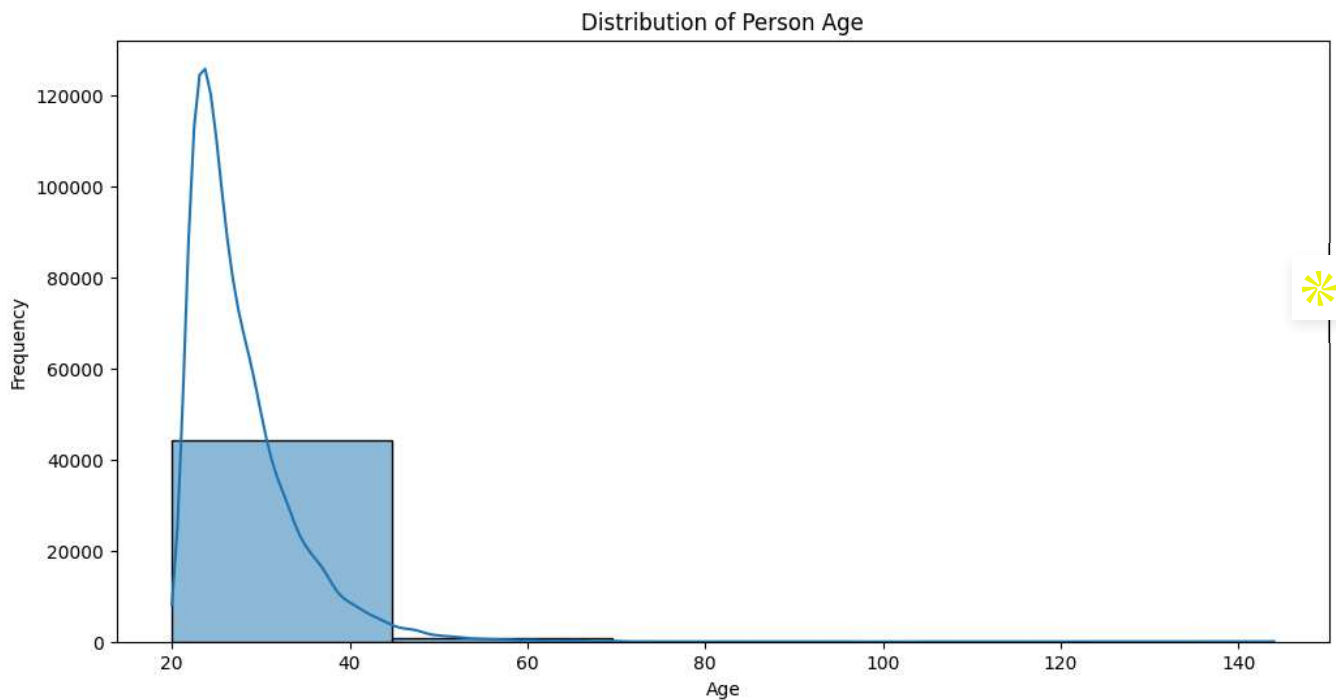
no_of_week_nights  type_of_meal_plan  required_car_parking_space  \
count      36275.000000    36275    36275.000000
unique         NaN         4         NaN
top         NaN     Meal Plan 1         NaN
freq         NaN    27835         NaN
mean     2.204300         NaN     0.030986
std     1.410905         NaN     0.173281
min     0.000000         NaN     0.000000
25%     1.000000         NaN     0.000000
50%     2.000000         NaN     0.000000
75%     3.000000         NaN     0.000000
max     17.000000         NaN     1.000000

room_type_reserved  lead_time  arrival_year  arrival_month  \
count      36275    36275.000000    36275.000000    36275.000000
unique         7         NaN         NaN         NaN
top     Room_Type 1         NaN         NaN         NaN
freq     28130         NaN         NaN         NaN
mean         NaN     85.232557    2017.820427     7.423653
std         NaN     85.930817     0.383836     3.069894
min         NaN     0.000000    2017.000000     1.000000
25%         NaN     17.000000    2018.000000     5.000000
50%         NaN     57.000000    2018.000000     8.000000
75%         NaN    126.000000    2018.000000    10.000000
max         NaN    443.000000    2018.000000    12.000000

arrival_date  market_segment_type  repeated_guest  \
count      36275.000000    36275    36275.000000
unique         NaN         5         NaN
top         NaN     Online         NaN
freq         NaN    23214         NaN
mean     15.596995         NaN     0.025637
std     8.740447         NaN     0.158053
min     1.000000         NaN     0.000000
25%     8.000000         NaN     0.000000
50%    16.000000         NaN     0.000000
75%    23.000000         NaN     0.000000
max    31.000000         NaN     1.000000
```

	no_of_previous_cancellations	no_of_previous_bookings_not_canceled	\
count	36275.000000	36275.000000	
unique	NaN	NaN	
top	NaN	NaN	
freq	NaN	NaN	

```
#distribution plots
plt.figure(figsize=(12,6))#size of plotting
sns.histplot(jaan['person_age'], kde=True, bins=5)
plt.title('Distribution of Person Age')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```



```
#loan defaults
default_rate=jaan.groupby("previous_loan_defaults_on_file")["loan_status"].mean()
print("\nLoan approval rate by previous defaults:\n", default_rate)
```

```
Loan approval rate by previous defaults:
previous_loan_defaults_on_file
No      0.45163
Yes     0.00000
Name: loan_status, dtype: float64
```

```
plt.figure(figsize=(12,6))
sns.histplot(jaan['credit_score'],kde=True, bins=5)
plt.title('Distribution of Credit Score')
```